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# Math Training — M1 Cognitive Science

## Session 3 — Analysis I

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Functions, derivatives, integrals, differential equations

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Today is about **functions of one variable**: what they are, how to measure the way they change, and how to add change up again.

- **Functions**: the handful of shapes that keep coming back.
- **Derivatives**: the slope, the rules, and what the *sign* tells you.
- **Integrals**: area under a curve, and the antiderivative.
- **A first differential equation**: what it means to *solve* one, and why the exponential is the answer.

Next session does the same for functions of *several* variables.

## ■ If you remember only one thing

A **derivative** says how fast something is changing right now. An **integral** adds those changes back up. A **differential equation** tells you the rate in terms of the thing itself — and to *solve* it is to produce the function that does what the equation says.

That last sentence is also a free check on every answer you will ever write: substitute it back and see whether both sides agree.

## ■ Definition

A **function**  $f$  is a rule that turns one number into another: give it  $t$ , it returns  $f(t)$ . Its **graph** is what you get by marking the height  $f(t)$  above each  $t$ .

Two words worth keeping straight:

- the **variable** is what you feed in — time, task difficulty, number of practice trials;
- the **parameters** are the numbers that fix *which* function you have — and they do not change as the variable runs.

In  $f(t) = a e^{-t/\tau}$ , the variable is  $t$ ;  $a$  and  $\tau$  are parameters.

# Five shapes cover almost everything

- **Constant**  $f(t) = c$  — a flat line. Nothing changes.
- **Linear**  $f(t) = mt + c$  — a straight line, climbing  $m$  per unit, starting at  $c$ .
- **Power**  $f(t) = t^n$  — and  $n$  may be negative or fractional:  $1/t = t^{-1}$ ,  $\sqrt{t} = t^{1/2}$ .
- **Exponential**  $f(t) = e^{at}$  — growth if  $a > 0$ , decay if  $a < 0$ . The shape this whole course runs on.
- **Logarithm**  $f(t) = \ln t$  — the inverse of the exponential: it undoes it.

The sine appears too, but only in Session 4.

## ■ Definition

$e \approx 2.718$  is the base for which the curve  $e^t$  climbs, at every point, at exactly the height it has reached. For  $a < 0$  write  $a = -1/\tau$ :

$$f(t) = e^{-t/\tau},$$

and the positive number  $\tau$  is the **time constant**.

## ■ Key idea

After one  $\tau$ , the quantity has fallen to  $e^{-1} \approx 37\%$  of where it started — and after another  $\tau$  to 37% of *that*, for ever.

Three numbers worth memorising:  $e^{-1} \approx 0.37$ ,  $e^{-2} \approx 0.14$ ,  $e^{-3} \approx 0.05$ . So “about three time constants” is the practical answer to “when is it over”.

## ■ Link to cognitive science

Learn a list of words, then test recall after a delay  $t$ . The proportion still recalled falls off as

$$R(t) = R_0 e^{-t/\tau},$$

the shape Ebbinghaus measured on himself in 1885 and the standard first description of forgetting ever since. Here  $R_0$  is what you recall immediately and  $\tau$  is how long the memory “lasts” in the sense above.

With  $R_0 = 0.9$  and  $\tau = 2$  days: after 2 days  $0.9 \times 0.37 \approx 0.33$ ; after 6 days  $0.9 \times 0.05 \approx 0.05$ . Two parameters, and the whole curve.

Real forgetting data are often fitted better by a power law  $R(t) = R_0 t^{-b}$  — worth one sentence, because it is a live debate and because it is a different one of our five shapes.

### ■ Practice — try it

1. Sketch  $f(t) = e^{-t/4}$  and  $g(t) = e^{-t}$  on the same axes. Which falls faster, and why?
2. Recall follows  $R(t) = 0.8e^{-t/3}$  with  $t$  in days. Give  $R(0)$ ,  $R(3)$ ,  $R(9)$ .
3. Simplify  $\ln(e^5)$ ,  $e^{\ln 7}$ ,  $\ln(a^2b)$ .
4. After how long has  $R$  fallen to half its initial value? (Solve  $e^{-t/\tau} = \frac{1}{2}$ .)



## ■ Definition

For a straight line, “how fast is it climbing” is the **slope**: rise over run. For a curve the slope changes from point to point, so we do the same thing *locally* — zoom in far enough and any smooth curve looks straight. The slope of that line is the **derivative**, written  $f'(t)$  or  $\frac{df}{dt}$ :

$$f'(t) = \lim_{h \rightarrow 0} \frac{f(t+h) - f(t)}{h}.$$

When the variable is time, the derivative is a *rate of change per unit time*. When it is anything else — difficulty, dose, number of trials — it is still “how much does the output move when I nudge the input”.

## ■ Useful fact

$$c \rightarrow 0, \quad t^n \rightarrow n t^{n-1}, \quad e^{at} \rightarrow a e^{at}, \quad \ln t \rightarrow \frac{1}{t}$$

$$(\alpha f + \beta g)' = \alpha f' + \beta g', \quad (fg)' = f'g + fg', \quad (f(g(t)))' = f'(g(t)) \cdot g'(t)$$

The power rule holds for negative and fractional  $n$  too:  $1/t \rightarrow -1/t^2$  and  $\sqrt{t} \rightarrow 1/(2\sqrt{t})$ .

The last one is the **chain rule**: differentiate the outside, then multiply by the derivative of the inside. It is where  $e^{at} \rightarrow a e^{at}$  actually comes from.

## ■ Watch out

$e^{-t/2}$  is *not* a power of  $t$ , so the power rule has nothing to say about it. Its derivative is  $-\frac{1}{2}e^{-t/2}$ .

Answers shaped like  $(-t/2)e^{(-t/2)-1}$  come from applying  $t^n \rightarrow nt^{n-1}$  to an exponential — treating the exponent as if it were the power.

The difference matters: in  $t^n$  the variable is in the *base*, in  $e^{at}$  it is in the *exponent*. Ask which rule applies before anyone writes anything down, and do it every time an exponential appears today.

$f' > 0 \Rightarrow$  rising,       $f' < 0 \Rightarrow$  falling,       $f' = 0 \Rightarrow$  momentarily flat

— a peak, a trough, or a value that has stopped moving.

## ■ Intuition

To find where a function is largest: set  $f' = 0$ , solve, and check the sign of  $f'$  on either side. Positive then negative means a maximum; negative then positive, a minimum.

Very often the sign is all you need, and you never have to evaluate anything.

## ■ Example

Differentiate  $f(t) = 3t^2 - 4t + 7$ ,  $R(t) = 0.9 e^{-t/2}$ ,  $RT(n) = 800 n^{-0.3}$ .

**Polynomial, term by term:**  $3t^2 \rightarrow 6t$ ,  $-4t \rightarrow -4$ ,  $7 \rightarrow 0$ , so  $f'(t) = 6t - 4$ . It vanishes at  $t = \frac{2}{3}$ , negative before and positive after: a minimum.

**Forgetting curve:**  $a = -\frac{1}{2}$ , so  $R'(t) = -\frac{1}{2} \cdot 0.9 e^{-t/2} = -\frac{1}{2} R(t)$ . Note what this says: *the rate of forgetting is proportional to how much you still remember* — that sentence is a differential equation, and we come back to it.

**Power law of practice:**  $RT'(n) = 800 \times (-0.3) n^{-1.3} < 0$  — reaction time keeps falling with practice, but the improvement per extra trial shrinks fast.

### ■ Practice — try it

1. Differentiate  $f(t) = t^2 - 4t + 7$ ,  $g(t) = 5e^{-2t}$ ,  $h(t) = 1/t + \sqrt{t}$ .
2. Differentiate  $R(t) = R_0 e^{-t/\tau}$  and show that  $R' = -R/\tau$ .
3. Use the chain rule on  $f(t) = e^{-t^2}$  and on  $g(t) = \ln(1 + 3t)$ .
4. Use the product rule on  $f(t) = t e^{-t}$ . Where is its maximum?

## ■ Definition

$\int_a^b f(t) dt$  is the **signed area** between the graph of  $f$  and the horizontal axis, from  $t = a$  to  $t = b$  — area above the axis counting positive, below it negative.

Two readings, both useful:

- chop  $[a, b]$  into thin slices of width  $dt$ , each contributing  $f(t) dt$ , and add them up. That is what the elongated S means: a sum.
- if  $f$  is a *rate*, the integral is the *total accumulated*. Speed integrates to distance; a rate of responding integrates to a number of responses.

## ■ Useful fact

If  $F' = f$ , then

$$\int_a^b f(t) dt = F(b) - F(a).$$

$F$  is called an **antiderivative** of  $f$ , and it is unique only up to an additive constant — hence the “ $+ C$ ” in the indefinite integral  $\int f(t) dt = F(t) + C$ .

So integration is differentiation run backwards, and the table of derivatives becomes a table of integrals read right to left:

$$\int t^n dt = \frac{t^{n+1}}{n+1} \quad (n \neq -1), \quad \int \frac{dt}{t} = \ln |t|, \quad \int e^{at} dt = \frac{e^{at}}{a}.$$



- **Substitution — the chain rule backwards**

$$\int f(g(t)) g'(t) dt = \int f(u) du \quad \text{with} \quad u = g(t), \quad du = g'(t) dt.$$

- **Integration by parts — the product rule backwards**

$$\int u v' dt = uv - \int u' v dt.$$

Choosing  $u$  in the second: pick the factor that gets *simpler* when differentiated. A polynomial times an exponential is the standard case — the polynomial is  $u$ , and each application drops its degree by one.

## ■ Example

Compute  $\int_0^T e^{-t/\tau} dt$ ,  $\int 2t e^{t^2} dt$ ,  $\int_0^\infty t e^{-t} dt$ .

**Directly:**  $\int_0^T e^{-t/\tau} dt = [-\tau e^{-t/\tau}]_0^T = \tau(1 - e^{-T/\tau})$ . As  $T \rightarrow \infty$  this tends to  $\tau$ : the total area under a decaying exponential is exactly its time constant.

**By substitution:**  $u = t^2$ ,  $du = 2t dt$ , so the integral is  $\int e^u du = e^{t^2} + C$ .

**By parts:**  $u = t$ ,  $v' = e^{-t}$ , so  $\int_0^\infty t e^{-t} dt = [-t e^{-t}]_0^\infty + \int_0^\infty e^{-t} dt = 0 + 1 = 1$ .

## ■ Practice — try it

1. Compute  $\int_0^2 (3t^2 - 4t + 1) dt$  and  $\int_1^e \frac{dt}{t}$ .
2. Compute  $\int_0^\infty e^{-3t} dt$ . Interpret the answer.
3. By substitution:  $\int \frac{2t}{1+t^2} dt$ .
4. By parts:  $\int t e^{-2t} dt$ .

## ■ Definition

If  $\dot{x} = f(t)$ , the rate depends on *time*, and you integrate:  $x(t) = \int_0^t f(s) ds + x_0$ . Nothing is circular — the right-hand side never mentions  $x$ .

If  $\dot{x} = f(x)$ , the rate depends on *the unknown itself*. You cannot integrate the right-hand side, because you do not yet know  $x(t)$  to integrate.

Only the second is a differential equation proper, and it needs a different move.

## ■ Watch out

This one is made by the *strong* students. Asked to solve  $\dot{x} = -(x - 3)$ , they anti-differentiate the right-hand side and return  $-x^2/2 + 3x$ . That is an antiderivative of something; it is not a solution of anything.

Two sentences fix it for good.

*First:* an antiderivative of a function of  $x$  is another function of  $x$ , but the answer you want is a function of  $t$  — the types do not even match.

*Second:* substitute and check. By the chain rule  $\frac{d}{dt}(-x^2/2 + 3x) = (-x + 3)\dot{x}$ , which equals  $-(x - 3)$  only if  $\dot{x} = -1$  — not what was asked.

## ■ Definition

$$\frac{dx}{dt} = ax + b \quad \implies \quad x(t) = x^* + (x_0 - x^*) e^{at}, \quad x^* = -\frac{b}{a} \quad (a \neq 0),$$

where  $x^*$  is the **steady state** (the rate vanishes there) and  $\tau = -1/a$  is the **time constant**.

Read it as a sentence rather than a formula: *start at  $x_0$ , end at  $x^*$ , and close the gap between them exponentially*. One  $\tau$  closes 63% of the gap; three close about 95%.

If  $a = 0$  the equation is just  $\dot{x} = b$ , there is no steady state, and  $x(t) = x_0 + bt$  marches off linearly for ever. That is why  $a \neq 0$  is written.

## ■ Intuition

The rate, plotted against  $x$ , is the straight line  $ax + b$ , and its slope is  $a$ . Perturb the steady state by  $\delta$ : then  $\dot{\delta} = a\delta$ , which decays if and only if  $a < 0$ .

$a < 0 \Rightarrow$  **stable**: the system returns,       $a > 0 \Rightarrow$  **unstable**: it runs away.

There is a picture that carries the same information without any algebra: mark  $x^*$  on a line and put an arrow at each side pointing the way the rate says. Arrows pointing in, stable. Arrows pointing out, unstable. It is called the **phase line**, and unlike the formula it still works when the rate is a curve.

## ■ Link to cognitive science

Performance  $P$  improves with practice, but not without limit: there is a ceiling  $P_{\max}$ , and the closer you get the slower you improve. Written as an equation,

$$\frac{dP}{dt} = k(P_{\max} - P),$$

which is  $\dot{x} = ax + b$  with  $a = -k$  and  $b = kP_{\max}$ , so  $P^* = P_{\max}$  and

$$P(t) = P_{\max} - (P_{\max} - P_0)e^{-kt}.$$

**With  $P_0 = 20$ ,  $P_{\max} = 100$ ,  $k = 0.1$  per session:** after 10 sessions

$P = 100 - 80 \times 0.37 \approx 70$ ; after 30,  $P \approx 96$ . The time constant is  $\tau = 1/k = 10$  sessions.

Note that this is the forgetting curve upside down — same equation, same exponential, opposite direction. That is not a coincidence: both say “the rate is proportional to the distance still to go”.



## ■ Definition

$$x(t + \Delta t) \approx x(t) + \Delta t \cdot \dot{x}(t).$$

In words: look up how fast you are moving, move at that speed for a short while, then look again.

The step pretends the rate is constant across the interval, so the error over a fixed duration is proportional to  $\Delta t$  — halve the step and you roughly halve the error.

## ■ Watch out

The step cannot be arbitrarily large: for  $\dot{x} = ax$  with  $a < 0$  the scheme *blows up* unless  $\Delta t < 2/|a| = 2\tau$ . And stability is not accuracy — staying below  $2\tau$  only stops the simulation exploding.

### ■ Practice — try it

1. Which are integration problems and which are differential equations? (i)  $\dot{x} = 3t^2$ ; (ii)  $\dot{x} = -2x$ ; (iii)  $\dot{x} = -2x + 5$ . Solve all three with  $x_0 = 1$ .
2. For  $\dot{x} = -x + 4$ : give  $x^*$  and  $\tau$ , decide stability, and describe the trajectory from  $x_0 = 10$  and from  $x_0 = 0$ .
3. Recall obeys  $\dot{R} = -R/3$  with  $R_0 = 0.8$  ( $t$  in days). Write  $R(t)$  and give  $R(6)$ .
4. Take one Euler step of  $\dot{x} = -x + 2$  from  $x = 0$  with  $\Delta t = 0.1$ . What is the largest step that would not blow up?

- A **function** is a rule with a variable and some parameters; five shapes cover almost everything, and  $e^{-t/\tau}$  is the one that matters most.
- The **derivative** is a slope. Sign: rising, falling, or flat. Rules: power, exponential, sum, product, chain.
- The **integral** is a signed area, and the fundamental theorem makes it differentiation run backwards. Substitution and by parts are the two techniques we need.
- $\dot{x} = ax + b$  starts at  $x_0$ , ends at  $x^* = -b/a$ , and closes the gap like  $e^{at}$ . *Solving* it is not *integrating* it, and stability is the sign of  $a$ .

Next session: functions of *several* variables — surfaces, partial derivatives, gradients — and functions whose output is a *vector*.